

# Modeling future trajectories of the HIV epidemic: Kenya

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Preliminary results

17 December 2024



World Health  
Organization

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## Objectives of the project

- Contribute to the evidence base on the package of interventions needed to ensuring declining and sustaining HIV incidence reductions beyond 95-95-95
- Contribute to understanding of focusing HIV testing and HIV prevention interventions to different subpopulations under fixed budgets
- Generalise evidence, where possible, and document process to enable repeated application in other settings/countries

# Approach

**Phase 1:** Projections of HIV incidence and mortality under continuation of current interventions (“status quo”)

**Phase 2:** Projections of impact on new infections of scaling up of interventions to country targets, individually and in combination

**Phase 3:** Economic analysis to identify  
- cost per infection averted, cost per DALY averted

# Three models

Goals

Optima HIV

HIV Synthesis

# Targets for Interventions

**ANNEXURE 3**  
MONITORING FRAMEWORK



NATIONAL SYNDEMIC DISEASES  
CONTROL COUNCIL

## National Multisectoral HIV Prevention Acceleration Plan 2023–2030



| Level of measurement |                                     | Indicator                                           | Disaggregation                    | Baseline (2022)                     | 2023 | 2024 | 2025 | 2026 | 2027 | 2028 | 2029 | 2030 | Data Source         | Frequency |
|----------------------|-------------------------------------|-----------------------------------------------------|-----------------------------------|-------------------------------------|------|------|------|------|------|------|------|------|---------------------|-----------|
|                      | Increase in awareness of HIV status | Proportion of PLHIV aware of HIV status             | Age, sex, population type, county | 95%-Total<br>90%-Male<br>94%-Female | 95%  | 96%  | 96%  | 97%  | 98%  | 99%  | 99%  | 100% | HIV Estimates/ IBBS | Annual    |
|                      | Increase in PrEP use                | Proportion of at risk populations initiated on PrEP | Age, sex, population type, county | 39%                                 | 44%  | 49%  | 54%  | 60%  | 65%  | 70%  | 75%  | 80%  | KHIS                | Monthly   |

| Intervention                                                                                                | Target                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| General <b>community testing</b> in adults (administered by community health workers)                       | Increase such testing by 50%                                                                                                                             |
| Recent <b>sexual risk-informed testing</b> (clinic-based) (index testing, testing in STI clinics) in adults | Increase such testing by 50%                                                                                                                             |
| <b>General wide distribution of self tests</b> in adults                                                    | Increase such testing by 50%                                                                                                                             |
| Recent sexual <b>risk-informed self-testing</b> (index testing) for adults                                  | Increase such testing by 50%                                                                                                                             |
| <b>General community testing in AGYW</b> not focused only on those with recent sexual risk                  | Increase such testing by 50%                                                                                                                             |
| <b>Behaviour change advice for AGYW</b> to reduce condomless sex and condom provision                       | Proportion of people using condoms at last sexual encounter = 95%                                                                                        |
| <b>VMMC</b> males aged 15+                                                                                  | Proportion circumcised in priority counties = 80%                                                                                                        |
| Increased <b>oral PrEP</b> / PEP support and uptake                                                         | Proportion of at-risk populations initiated on (any) PrEP = 80%                                                                                          |
| Increased <b>oral PrEP</b> / PEP support and uptake <b>plus LA injectable</b>                               | Proportion of at-risk populations initiated on (any) PrEP = 80%                                                                                          |
| Increased <b>oral PrEP</b> / PEP support and uptake <b>plus LA injectable / DPV ring</b>                    | Proportion of at-risk populations initiated on (any) PrEP = 80%                                                                                          |
| <b>Harm reduction services</b> for PWID                                                                     | Proportion of PWID using sterile needles and syringes in the last injecting episode = 100%;<br>Proportion of PWID using Medically Assisted Therapy = 80% |
| Interventions to <b>maximize linkage to care and retention</b> in PLHIV                                     | Proportion of PLHIV with known HIV status on ART = 95%                                                                                                   |
| Interventions to <b>enhance adherence in people on ART</b>                                                  | Proportion of PLHIV on ART achieving viral suppression = 95%                                                                                             |
| Interventions to <b>maximize ART coverage in pregnant and breastfeeding women</b> living with HIV           | Proportion of PLHIV aware of HIV status = 100%<br>Increase testing by 50% ;<br>Proportion of PLHIV with known HIV status on ART = 95%                    |
| Maximize access to the <b>WHO AHD package of care</b>                                                       | WHO package of care fully implemented.                                                                                                                   |

Testing interventions

Primary prevention interventions

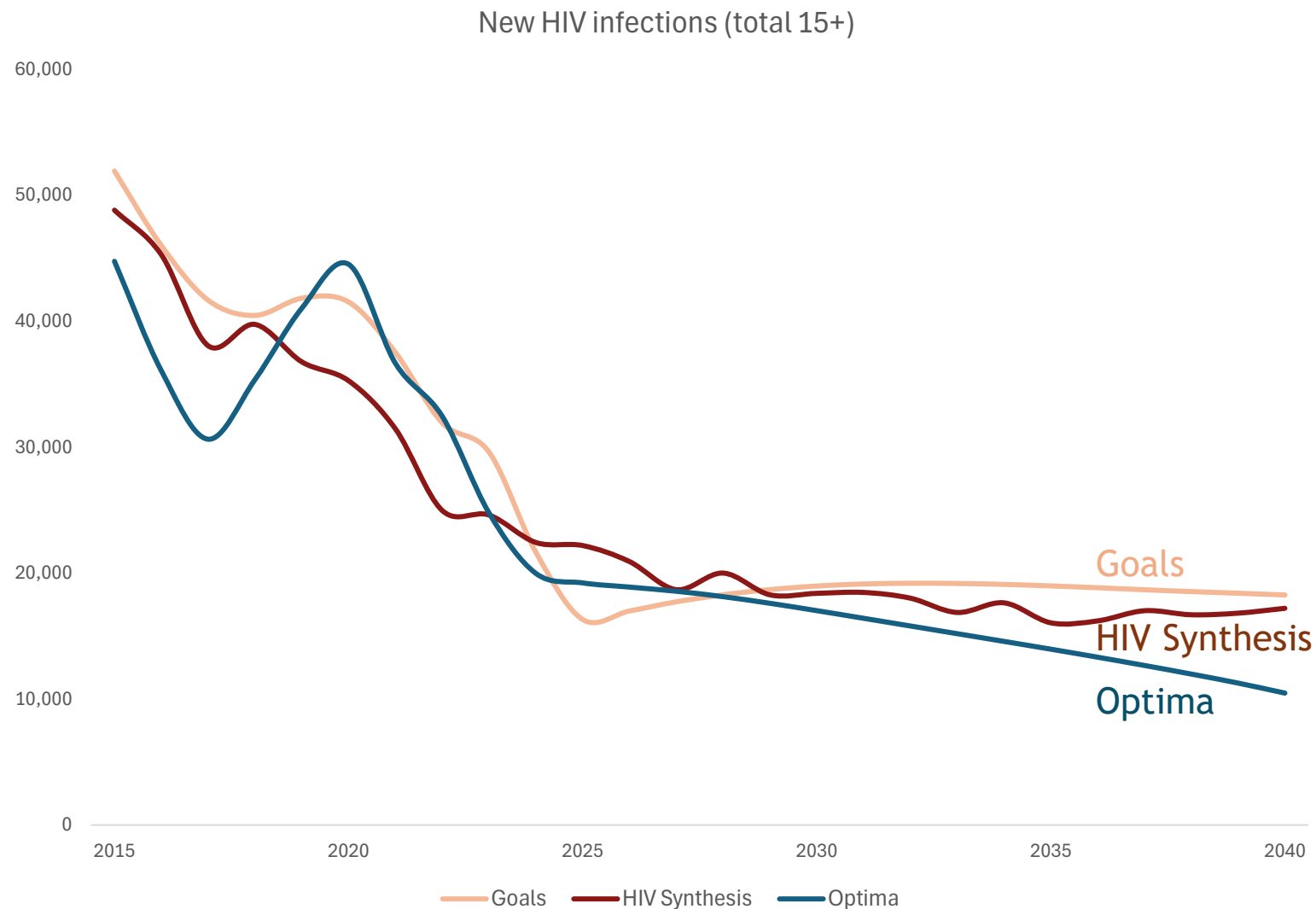
Treatment interventions

# Intervention scenarios

- Status Quo: Current coverage (2023) continued with no scale up
- Effect of scaling specific interventions to the national target in 2030 compared to status quo
  - We did not assess feasibility of ambitious country targets
  - Differences in model structure mean that different models may have implemented different methods of intervention scale up to reach targets
- All intervention targets achieved by 2030
- Mid-way to intervention targets reached by 2030
  - Individual interventions
  - All interventions together

# HIV incidence among 15+

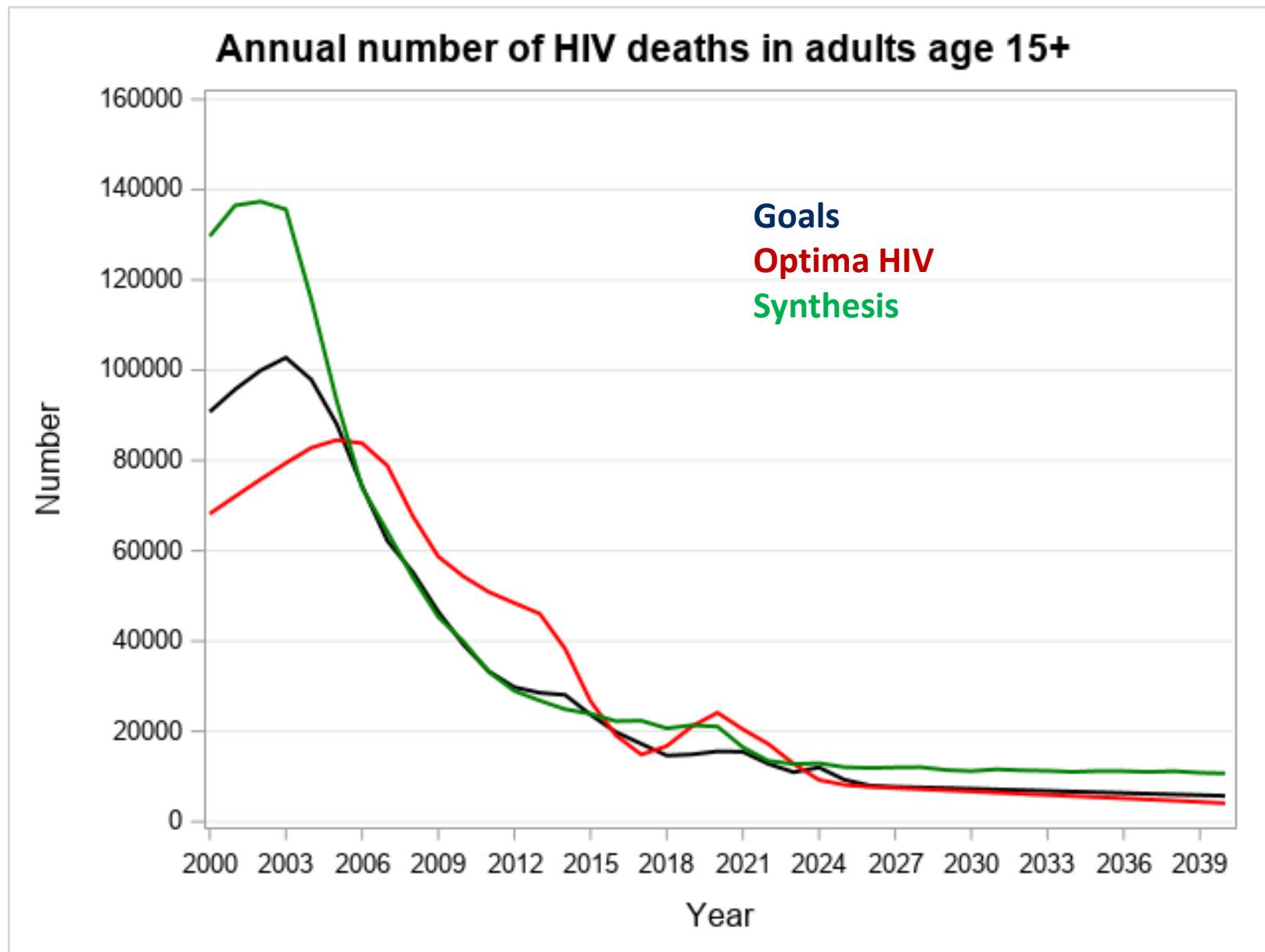
3 models' projections to 2040 under Status Quo





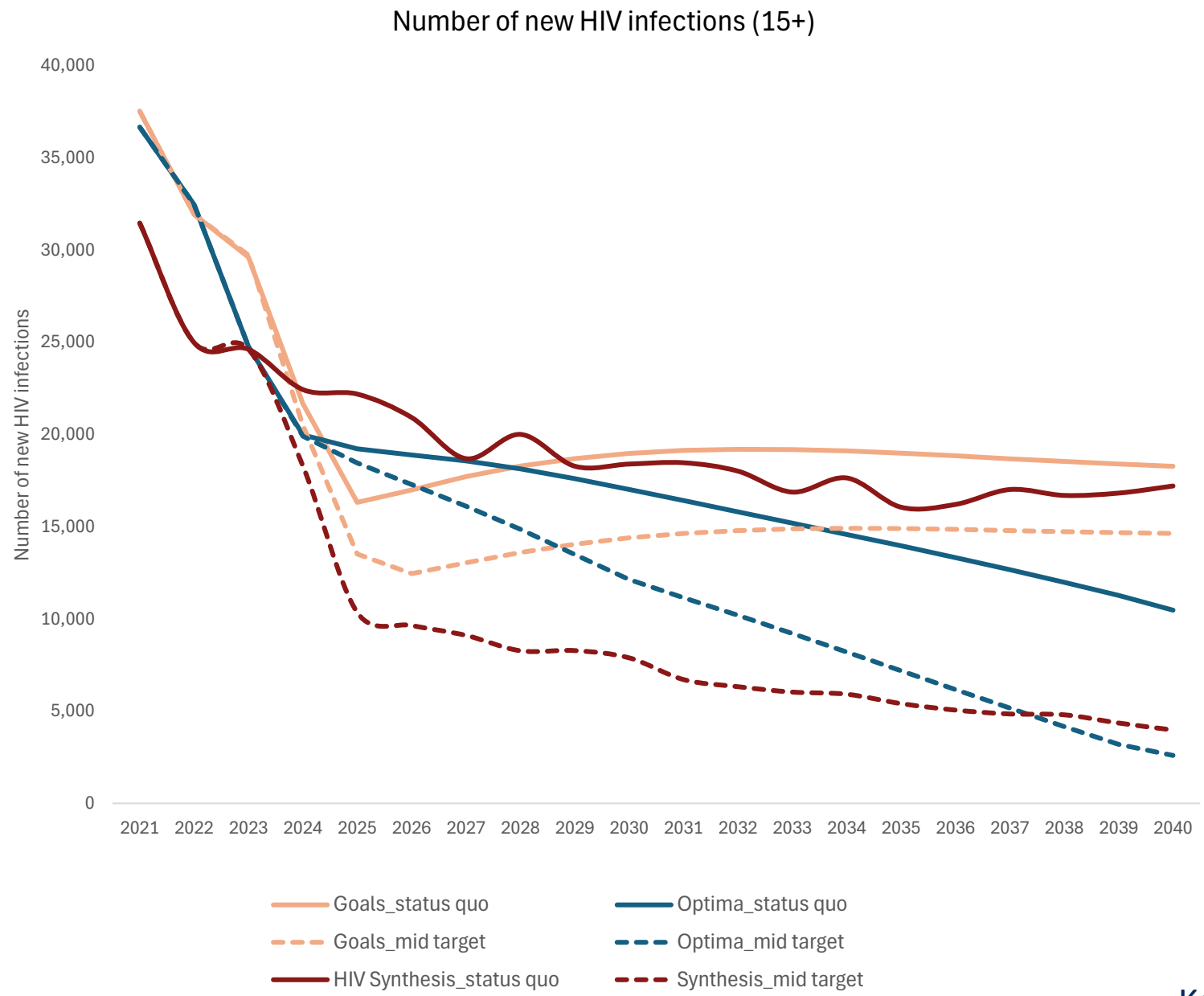
# HIV deaths among 15+

Status quo scenario:  
continuation of  
interventions as they are



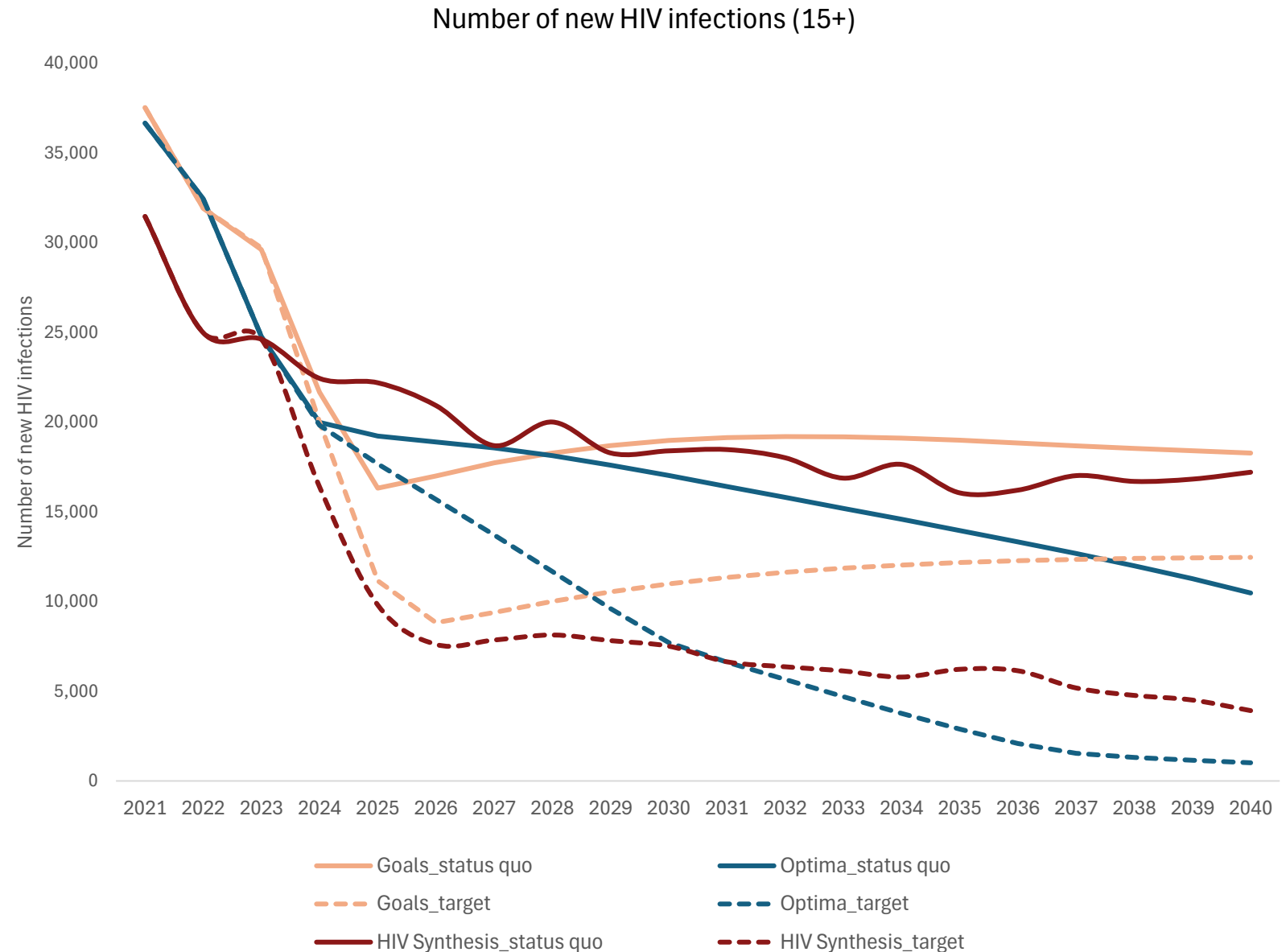
# Effect of Interventions on new HIV infections

All interventions scaled up to midway targets by 2030 could avert 57,600 - 85,600 (18%-32%) over the 2024-2040 period

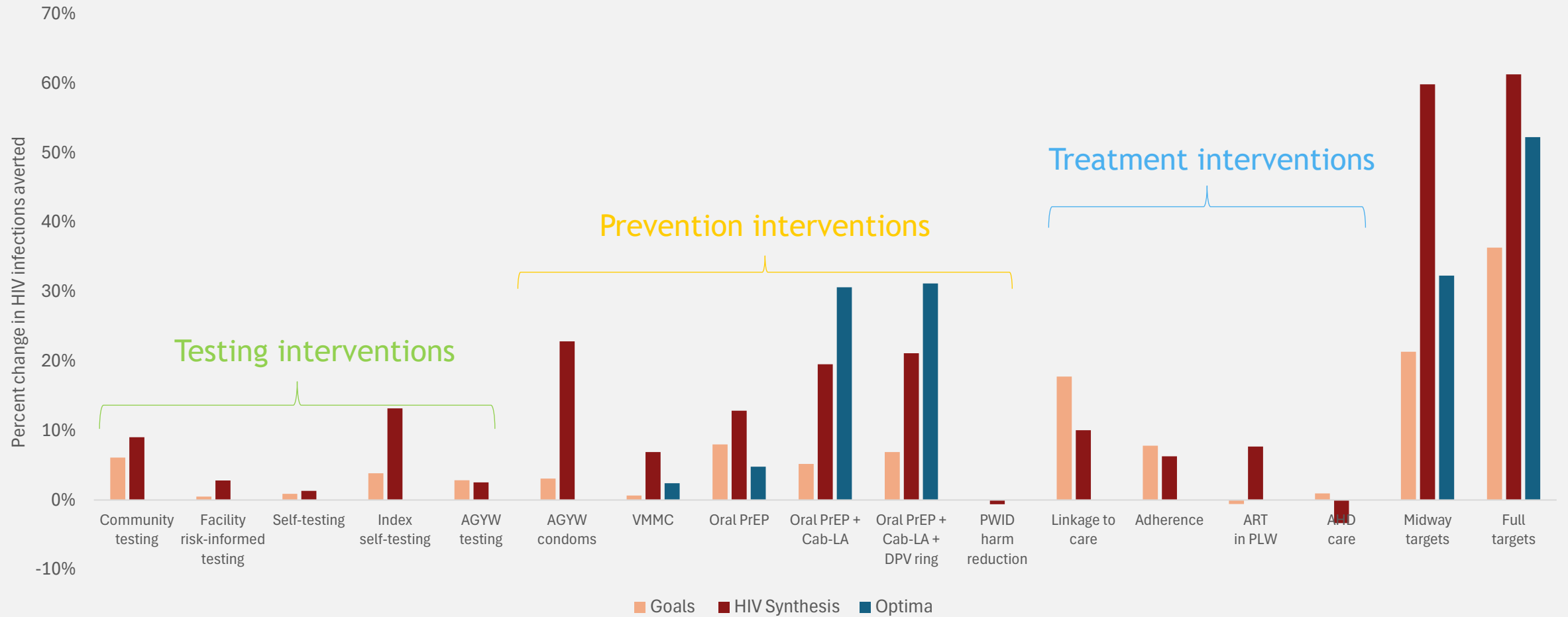


# Effect of Interventions on new HIV infections

All interventions scaled up to **full** targets by 2030 could avert 115,200 – 191,200 (36%-61%) over the 2024-2040 period



## Proportional change new HIV infections compared with SQ



# Incremental cost per DALY averted



No range around geometric mean: only one model included this intervention

# Summary of key findings

- Achieving targets for all interventions will lead to substantial and sustained declines in HIV incidence
  - Note that some differences between models relate to having modelled different intervention scales, including due to differences in model structures
- ICERs for ART retention and advanced HIV disease package of care suggest enhanced implementation of these interventions is cost-effective, but further investigation, including confirmation of actual costs of further scale up of these interventions, needed before making firm conclusions
- Index self-testing showed impact; wide range of ICERs for AGYW testing
- Among prevention interventions, oral PrEP had the lowest cost/infection averted; harm reduction had the highest

# Thank you

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